Mentor Dashboard

Authentication & Profile Management

1. Mentor Registration

This is where a new mentor signs up for the platform.

- **Email:** A text box where the mentor types their email address.
- **Password:** A text box for creating a password, often with a "Confirm Password" box to ensure it's typed correctly.
- **First Name, Last Name:** Simple text boxes for basic identification.
- **Phone Number:** A text box for their contact number.
- **Primary Skill/Area of Expertise:** This could be a text box for free-typing or a dropdown menu to select from a predefined list of skills (e.g., "Software Development," "Business Strategy").
- **"I agree to Terms & Conditions":** A **checkbox** that must be ticked before proceeding.
- **"Sign Up" / "Register" Button:** A clickable button to submit the registration form.

2. Professional Profile

After registration, mentors build out their detailed professional background.

- **Bio/About Me:** A large **text area** where mentors can write a comprehensive description of themselves, their experience, and their approach.
- **Work Experience:** This would typically involve multiple sets of input fields:
  - **Job Title:** Text box.
  - **Company Name:** Text box.
  - **Start Date / End Date:** Date pickers (calendar pop-ups) or dropdowns for month and year.
  - **Job Description:** A text area for responsibilities and achievements.
  - An **"Add New Experience" button** to add more entries.
- **Certifications:** Similar to work experience, fields for:
  - **Certification Name:** Text box.
  - **Issuing Body:** Text box.
  - **Date Obtained:** Date picker or dropdowns.
  - An **"Add New Certification" button**.
- **Specialization Areas/Skills:** This often uses:
  - A **multi-select dropdown** where mentors can pick multiple skills from a list.

        - Or, a text box that allows typing skills, which then convert into **tags** (like little labels).
    - **LinkedIn Profile URL:** A text box for their LinkedIn profile link.
    - **"Save Profile" Button:** To save all the entered information.

3. Skill Assessment

This feature evaluates a mentor's proficiency in their stated skills.

- **Multiple Choice Questions:** Each question would have **radio buttons** (for a single correct answer) or **checkboxes** (for multiple correct answers).
- **Short Answer/Essay Questions: Text areas** for written responses.
- **Coding Challenges:** For technical skills, there might be a specialized **code editor** embedded in the page where mentors can write and sometimes even run code.
- **"Start Assessment" / "Submit Assessment" Buttons:** To begin and finish the test.

4. Document Upload

Mentors upload important files like resumes and certificates.

- **"Upload Resume" Button:** A standard **file upload button** that, when clicked, opens your computer's file browser.
- **"Upload Certificate" Button:** Another file upload button, possibly labeled specifically for certificates.
- **"Upload Professional Credentials" Button:** Similar file upload for other relevant documents.
- Sometimes, after uploading, there might be a **dropdown** to categorize the document (e.g., "Degree," "License").

5. Availability Management

Mentors set their working hours and when they are free.

- **Input Fields:**
    - **Time Zone Selector:** A **dropdown menu** to ensure their availability is shown correctly to people in different parts of the world.
    - **Working Hours Selector (for each day of the week):** This could be a combination of **dropdowns** for hours and minutes (e.g., "From: 9:00 AM To: 5:00 PM") or sliders.
    - **Availability Calendar:** A visual calendar where mentors can click on specific dates or drag across time slots to mark them as available or unavailable.
    - **"Add Block Out Time" Button:** To specify periods they are unavailable, like vacations, using date and time pickers.
    - **"Save Availability" Button:** To update their schedule.

6. Profile Verification

This is an administrative process to verify the mentor's credentials.

- **"Approve Profile" / "Reject Profile" Buttons:** Simple, clear buttons on the mentor's profile review page.
- **Reason for Rejection:** If rejecting, a **text area** usually appears, requiring the admin to provide an explanation.
- **Verification Status:** A **dropdown menu** or set of **radio buttons** to change the mentor's status (e.g., "Pending Review," "Verified," "Rejected").
- **Internal Notes:** A **text area** for the verification team to add private comments.

7. Rating System

Mentees can provide feedback on mentors.

- **Star Rating:** A row of **5 stars** that a mentee clicks to select a rating (e.g., clicking the third star gives a 3-star rating). ⭐⭐⭐
- **Review Text:** A **text area** where the mentee can write their detailed feedback.
- **"Submit Review" Button:** To post the rating and comments.

8. Professional Networking

Mentors can connect with other professionals on the platform.

- **Search Bar:** A **text box** where mentors can type names, skills, or specializations to find other mentors.
- **"Connect" / "Send Connection Request" Button:** A clickable button that appears on another mentor's profile, sending a request to connect.
- **"Accept" / "Decline" Buttons:** Buttons that appear on a mentor's "Connection Requests" page to manage incoming requests.

Mentor Dashboard Overview

1. Student Overview

This section provides a quick look at assigned students and their progress.

- **Search Bar:** A **text box** to type a student's name and quickly find them.
- **Filter Options: Dropdown menus** or **checkboxes** to filter students by status (e.g., "Active," "Completed," "At Risk") or by topic.
- **Sort By: Dropdown menu** to sort the student list (e.g., by "Name," "Progress," "Last Activity").
- **"View Details" Button:** A clickable button next to each student's name to go to their individual profile or progress page.

2. Task Management Summary

This summarizes pending tasks, submissions, and evaluations.

- **Filter by Status: Dropdown menu** or **radio buttons** to show "Pending Tasks," "Pending Submissions," or "Pending Evaluations."
- **Filter by Due Date: Dropdown menu** (e.g., "Today," "Next 7 Days," "Overdue") or a **date range picker** to narrow down tasks.
- **"Mark as Complete" Checkbox/Button:** Next to each task or submission, a **checkbox** or a clickable button to mark it as done after review.
- **"Review Submission" Button:** A clickable button to open a student's submitted work for evaluation.
- **"Add New Task" Button:** To create a new task for a student or yourself.

3. Performance Metrics

This displays data on mentor effectiveness and student success rates.

- **Time Period Selector: Dropdown menus** or **radio buttons** to view metrics for "Last 30 Days," "This Quarter," "All Time," etc.
- **"View Report" Button:** A button that might generate a more detailed report or open a specific analytics page.
- **Metric Selection: Checkboxes** or **toggles** to choose which specific metrics to display (e.g., "Student Completion Rate," "Average Rating," "Response Time").

4. Schedule Management

An overview of daily, weekly, and monthly schedules.

- **Input Fields:**
  - **Date Navigator: Left/Right arrow buttons** to move between days, weeks, or months, and a **calendar icon** to jump to a specific date.

- **View Selector: Buttons** or **tabs** to switch between "Daily," "Weekly," and "Monthly" views.
- **"Add Event/Appointment" Button:** To open a form for scheduling a new meeting or event (with input fields for date, time, attendees, topic).
- **"Edit" / "Delete" Icons:** Small clickable icons next to existing schedule entries to modify or remove them.

5. Notification Center

Provides real-time notifications and alerts.

- **"Mark as Read" Checkbox/Button:** Next to each notification, to dismiss it.
- **"Clear All Notifications" Button:** To clear all visible alerts.
- **Notification Settings (gear icon):** A clickable icon that leads to a page with **checkboxes** or **toggles** to customize which types of notifications are received (e.g., "Email me about new messages," "Show pop-up for new submissions").

6. Quick Actions

Shortcuts to frequently used functions.

- **Action Buttons:** Clearly labeled **buttons** for common tasks like "Send Message to Student," "Schedule a Session," "Create New Task," etc. These buttons act as shortcuts, often opening new forms that contain their own sets of input fields.
- **"Customize Quick Actions" Button:** A button that allows mentors to select or reorder which quick actions appear on their dashboard using **checkboxes** or **drag-and-drop** interfaces.

7. Activity Timeline

Shows recent activities and interactions.

- **Filter by Type: Dropdown menu** or **checkboxes** to show specific types of activities
- **Filter by Student:** A **dropdown menu** to view activities related to a particular student.
- **"Load More" Button:** To display older activities on the timeline.

8. Goal Tracking

Monitors personal and student goal progress.

- **Goal Status Update: Dropdown menus** or **sliders** next to each goal to update its progress (e.g., "Not Started," "In Progress," "Completed," or a percentage).
- **"Add New Goal" Button:** To open a form for setting a new goal (with input fields for goal description, target date, associated student).
- **"Edit" / "Delete" Icons:** To modify or remove existing goals.
- **Goal Filter: Dropdown menu** to filter goals by "Personal," "Student Specific," "Active," "Completed."

Student Assignment & Management System

1. Student Allocation

This feature allows assigning students to mentors, either automatically or manually.

- **"Allocate Automatically" Button:** A clickable button to trigger the system's automated assignment process (which might use the "Advanced matching algorithms" mentioned).
- **Manual Assignment:**
  - **Student Selector:** A **dropdown menu** or **searchable list** to select one or more students.
  - **Mentor Selector:** A **dropdown menu** or **searchable list** to choose a specific mentor.
  - **"Assign" Button:** A button to confirm the manual assignment.
- **Assignment Rules Configuration:** (For automated allocation) **Text fields**, **dropdowns**, or **checkboxes** to set parameters like "prioritize skill match," "balance mentor workload," or "assign to specific groups."

2. Batch Management

This allows managing groups of students at once.

- **"Create New Batch" Button:** To open a form for defining a new batch.
  - **Batch Name:** A **text box** for the batch's name (e.g., "Fall 2025 Intake").
  - **Start Date / End Date: Date pickers** to define the batch's duration.
  - **Enrollment Criteria: Dropdowns** or **checkboxes** to specify which students belong to this batch (e.g., "All new registrations," "Students from a specific course").
- **Batch Selector:** A **dropdown menu** to choose an existing batch to manage.
- **"Add Students to Batch" Button:** To select students (using a **searchable list** or **checkboxes**) and assign them to the current batch.
- **"Remove Students from Batch" Button:** To remove selected students from a batch.
- **"Archive Batch" / "Delete Batch" Buttons:** To manage the lifecycle of a batch.

3. Student Profiles

This provides a detailed view of each student. While primarily for display, there are input fields for updating information.

- **Input Fields (for administrators/mentors to update information):**
  - **Search Bar:** A **text box** to find a student by name or ID.

- **Edit Profile Button:** A clickable button on a student's profile page that turns display fields into editable **text boxes**, **dropdowns**, and **checkboxes** for details like contact info, academic background, or interests.
- **"Upload Student Documents" Button:** A **file upload button** for resumes, transcripts, etc.
- **"Save Changes" Button:** To commit any updates made to the profile.

4. Skill Matching

This feature helps match students to mentors based on their skills and interests.

- **Input Fields:**
  - **Skill Tags (for students):** When a student creates their profile, they'd use **text fields** to enter skills (which might convert to tags) or select from a **multi-select dropdown**.
  - **Interest Selection (for students):** Similar to skills, **checkboxes** or a **multi-select dropdown** for areas of interest.
  - **Skill Search (for administrators/mentors):** A **text box** to search for students or mentors with specific skills.
  - **Match Criteria Settings:** (For automated matching) **Sliders** or **text fields** to adjust the weight of different matching criteria (e.g., "Priority on technical skills: 80%, Interests: 20%").
  - **"Run Skill Match" Button:** To initiate a search or re-evaluation based on skill compatibility.

5. Capacity Management

This controls the maximum number of students a mentor can be assigned.

- **Mentor Selector:** A **dropdown menu** to select a specific mentor.
- **Maximum Student Count:** A **number input field** (with up/down arrows) to set the cap for that mentor.
- **"Set Global Capacity" Field:** A **number input field** to set a default maximum for all new mentors.
- **"Apply Capacity" Button:** To save the changes to mentor capacities.

6. Transfer Requests

This handles requests for students to be reassigned to a different mentor.

- **Input Fields (for students or administrators submitting a request):**
  - **Student Selector:** A **dropdown menu** to select the student for whom the transfer is requested.
  - **Current Mentor (auto-filled):** This would likely be displayed automatically.
  - **Requested New Mentor Selector:** A **dropdown menu** to choose the desired new mentor (if known).

    - **Reason for Transfer:** A **text area** where the user explains why the transfer is needed.
    - **"Submit Transfer Request" Button:** To send the request.
- **Input Fields (for administrators approving/denying requests):**
    - **"Approve" / "Deny" Buttons:** Clickable buttons on the request review page.
    - **Comments Field:** A **text area** for the administrator to add notes for approval or denial.

7. Group Formation

This allows creating and managing student groups for collaborative projects.

- **"Create New Group" Button:** To open a form for defining a new group.
    - **Group Name:** A **text box** for the group's name.
    - **Project/Topic:** A **text box** or **dropdown** for the group's focus.
    - **Student Selector:** A **multi-select list** or **checkboxes** to add students to the group.
    - **Mentor (Optional):** A **dropdown** to assign a mentor specifically to this group.
- **"Add/Remove Students from Group" Buttons:** To modify existing groups using student selectors.
- **"Disband Group" Button:** To remove a group.

8. Individual Tracking

This monitors each student's progress and performance.

- **Student Selector/Search:** A **dropdown menu** or **search bar** to quickly jump to a specific student's tracking overview.
- **Progress Update (for mentors): Sliders**, **percentage input fields**, or **status dropdowns** (e.g., "On Track," "Behind Schedule") to update a student's overall progress.
- **Feedback Input:** A **text area** for mentors to provide written feedback on performance, with a **"Submit Feedback" button**.
- **Goal Progress Updates:** (If integrated with a goal tracking system) **Checkboxes** or **progress bars** that can be interacted with to mark goal completion.
- **"Log Activity" Button:** To add a record of an interaction with the student (e.g., "Met for 1:1 session"). This would open a form with fields for date, time, and notes.

Comprehensive Task Management System

1. Task Creation Interface

This is where users define new tasks.

- **Task Title:** A **text box** for a concise name of the task (e.g., "Complete Research Paper," "Develop Login Module").
- **Task Description:** A **rich text editor** (like a mini-word processor within the system) for detailed instructions, formatting, links, and even embedded images.
- **Assigned To:** A **dropdown menu** or a **searchable list** to select the student(s) or team members responsible.
- **Due Date:** A **date picker** (a calendar pop-up) to select the deadline.
- **Priority:** A **dropdown menu** (e.g., "High," "Medium," "Low") or a set of **radio buttons** to indicate urgency.
- **Category:** A **dropdown menu** or **multi-select checkboxes** to categorize the task (e.g., "Coding," "Research," "Presentation").
- **"Create Task" Button:** A clickable button to save and create the task.

2. Task Templates

This feature allows users to quickly create tasks using pre-defined structures.

- **Template Selector:** A **dropdown menu** or a list of clickable **template cards** to choose a pre-built task structure
- **"Apply Template" Button:** After selecting, this button populates the "Task Creation Interface" with the template's details. Users can then modify these pre-filled fields as needed.
- **"Save as Template" Button:** On a newly created or edited task, a button that allows saving its structure as a new template (requires a **text box** for the template name).

3. Multi-Type Tasks

The system supports different kinds of tasks (coding, research, etc.).

- **Task Type Selector:** This is often a **dropdown menu** or a set of **radio buttons/icons** during task creation (e.g., "Coding Task," "Research Task," "Presentation Task," "Project Task"). Selecting a type might change the visibility of other relevant input fields (e.g., a "Coding Task" might show a field for "Programming Language").
- **Specific Input Fields based on Type:**
  - For **Coding:** A **text box** for a link to a code repository, or even an embedded code editor.
  - For **Research:** A **text box** for expected sources or word count.
  - For **Presentation:** A **text box** for presentation length or specific topics to cover.

4. Task Categories

Tasks can be organized by skill area, difficulty, and project phase.

- **Category Selector:** During task creation or editing, a **dropdown menu** or **multi-select checkboxes** for selecting pre-defined categories (e.g., "Frontend Development," "Data Analysis," "User Interface Design").
- **Difficulty Level:** A **dropdown menu** (e.g., "Easy," "Medium," "Hard") or a **slider**.
- **Project Phase:** A **dropdown menu** (e.g., "Planning," "Development," "Testing," "Deployment").
- **"Manage Categories" Button (for administrators/mentors):** Leads to a page with **text boxes** and **"Add/Edit/Delete" buttons** for creating and managing the list of available categories.

5. Deadline Management

Setting deadlines and handling reminders/escalations.

- **Due Date:** A **date picker** and a **time picker** (dropdowns for hour/minute) for precise deadlines during task creation.
- **Reminder Settings:**
  - **"Send Reminder" Checkbox:** To enable reminders.
  - **Reminder Frequency:** A **dropdown menu** (e.g., "1 Day Before," "3 Hours Before," "Custom").
  - **Custom Reminder Time:** If "Custom" is selected, **number input fields** for "Days before" or "Hours before."
- **Escalation Rule:** A **dropdown menu** to select who should be notified if a task is overdue (e.g., "Mentor," "Project Manager").

6. Priority System

Tasks are visually prioritized.

- **Priority Level:** A **dropdown menu** (e.g., "Urgent," "High," "Medium," "Low") or **radio buttons** during task creation/editing. These selections directly influence the visual indicators on the Kanban board (e.g., color coding, icon).
- **"Sort by Priority" Button/Dropdown:** On the task list or Kanban board, a clickable option to reorder tasks based on their priority level.

7. Bulk Operations

Creating, assigning, and managing multiple tasks at once.

- **Task Selection (Checkboxes): Checkboxes** next to each task in a list view, allowing multiple tasks to be selected simultaneously.
- **Bulk Action Dropdown:** After selecting tasks, a **dropdown menu** appears with options like "Assign to," "Change Status," "Set Due Date," "Delete Selected."
- **Target Input Fields (after selecting an action):** Depending on the action chosen from the dropdown, new input fields appear (e.g., a "Mentor Selector" for "Assign to," a "Status Dropdown" for "Change Status").
- **"Apply" Button:** To execute the bulk operation.

8. Task Dependencies

Setting prerequisites for tasks.

- **"Add Dependency" Button:** During task creation or editing, a button that opens a new section.
- **Dependent Task Selector:** A **searchable list** or **dropdown menu** to choose other tasks that must be completed *before* the current task can start.
- **Dependency Type:** A **dropdown menu** (e.g., "Finish-to-Start," "Start-to-Start," "Finish-to-Finish") if the system supports complex dependencies.
- **"Remove Dependency" Icon:** A small "X" icon next to each listed dependency to remove it.

9. Collaborative Tasks

Managing group tasks with roles and collaboration tools.

- **"Group Task" Checkbox:** A checkbox during task creation to designate it as a group task.
- **Team/Group Selector:** If checked, a **dropdown menu** or **searchable list** to select the specific student group assigned.
- **Role Assignment:** For each team member on the task, a **dropdown menu** to assign specific roles (e.g., "Lead," "Researcher," "Developer," "Presenter").
- **Shared Document Link:** A **text box** to paste a link to a shared document (Google Docs, etc.).
- **Discussion Forum Link:** A **text box** to link to a dedicated discussion thread within the system or an external tool.
- **"Start Collaboration" Button:** Might open an integrated communication tool or link to one.

Advanced Progress Tracking & Analytics

1. Real-time Progress Dashboard

This provides live updates on student progress.

- **Student Selector:** A **dropdown menu** or **search bar** to quickly choose a specific student whose live progress you want to view.
- **Task/Project Filter: Dropdowns** or **checkboxes** to filter the dashboard to show progress on specific tasks, projects, or courses.
- **Time Period Selector: Dropdowns** like "Today," "This Week," or a **date range picker** to narrow down the displayed activity.
- **"Refresh" Button (or auto-refresh toggle):** A button to manually update the data, or a **toggle switch** to enable/disable automatic real-time updates.

2. Performance Analytics

This provides detailed insights into student performance trends.

- **Metric Selector: Checkboxes** or a **multi-select dropdown** to choose which performance metrics to analyze (e.g., "Assignment Scores," "Completion Rates," "Time Spent on Tasks").
- **Timeframe Selector: Dropdowns** like "Last 30 Days," "Last Quarter," "Custom Date Range" (with a **date range picker**), to define the period for analysis.
- **Group/Batch Filter:** A **dropdown menu** to analyze performance for a specific student group or batch.
- **Visualization Type: Radio buttons** or **icons** to switch between different chart types (e.g., "Line Chart," "Bar Graph," "Heatmap") if multiple options are available.
- **"Generate Report" / "View Analytics" Button:** To apply the selected filters and display the data.

3. Skill Development Tracking

This monitors how a student's skills improve over time.

- **Student Selector:** A **dropdown menu** or **search bar** to pick a student.
- **Skill Selector:** A **multi-select dropdown** to choose specific skills to track (e.g., "Problem Solving," "Python Programming," "Communication").
- **"Compare With" Selector:** A **dropdown menu** to compare a student's skill development against the class average or a previous period.
- **"View Skill Trend" Button:** To display a chart showing skill improvement.

4. Milestone Tracking

This tracks the completion of internship phase milestones.

- **Internship Phase Selector:** A **dropdown menu** to view milestones for a specific phase (e.g., "Phase 1: Onboarding," "Phase 2: Project Initiation").
- **Student Filter:** A **dropdown menu** or **search bar** to view milestone progress for individual students.
- **Milestone Status Updater (for mentors/admins):** On a milestone checklist, **checkboxes** or **status dropdowns** (e.g., "Not Started," "In Progress," "Completed") to update the status of each milestone.
- **"Add New Milestone" Button:** To open a form for defining a new milestone (with **text fields** for name, description, and target date).

5. Comparative Analysis

This allows comparing student performance within batches.

- **Batch Selector:** A **dropdown menu** to select the batch you want to analyze.
- **Student Selector (for comparison): Checkboxes** or a **multi-select list** to pick specific students within that batch to compare side-by-side.
- **Performance Metric Selector: Dropdown menu** to choose the specific metric for comparison (e.g., "Average Score," "Time to Complete," "Engagement Level").
- **"Run Comparison" Button:** To generate the comparative report or visualization.

6. Predictive Analytics

This uses AI to forecast student success.

- **Prediction Scope: Radio buttons** or a **dropdown menu** to select what you want predictions for (e.g., "Risk of Dropping Out," "Likelihood of Project Completion," "Future Skill Proficiency").
- **Student/Group Filter:** A **dropdown menu** to narrow down predictions to a specific student or group.
- **"Generate Prediction" Button:** To display the AI-powered insights and recommendations.
- **Confidence Threshold:** (Advanced) A **slider** to set the confidence level for predictions.

7. Custom Metrics

This allows defining and tracking personalized performance indicators.

- **"Create New Metric" Button:** To open a form for defining a custom metric.
  - **Metric Name:** A **text box** (e.g., "Active Learning Hours," "Collaboration Score").
  - **Calculation Formula/Criteria:** A **text area** or **guided builder** to define how this metric is calculated (e.g., "Sum of video watch time + 2x forum posts").
  - **Target Value:** A **number input field** to set a goal for the metric.
  - **Visibility: Checkboxes** to select who can see this metric (e.g., "Mentors Only," "Students and Mentors").
- **"Manage Custom Metrics" Button:** Leads to a list of existing custom metrics where users can **edit** or **delete** them.

8. Progress Reports

This feature automates the generation of progress reports.

- **Student Selector:** A **dropdown menu** or **searchable list** to select the student(s) for whom to generate a report.
- **Report Template Selector:** A **dropdown menu** to choose a specific report format (e.g., "Weekly Summary," "Detailed Monthly Report," "Final Performance Review").
- **Date Range Selector:** A **date range picker** to define the period the report should cover.
- **Include Sections: Checkboxes** to select which specific sections to include in the report (e.g., "Skill Progress," "Task Completion," "Mentor Feedback").
- **Output Format: Radio buttons** or a **dropdown menu** to choose the file type (e.g., "PDF," "CSV," "Email").
- **"Generate Report" Button:** To create and download/send the report.

9. Visual Progress Maps

This provides interactive visualizations of student learning journeys.

- **Student Selector:** A **dropdown menu** to view a specific student's learning map.
- **Zoom/Pan Controls: Buttons** or **mouse/touch gestures** (not typical input fields, but interactive elements) to navigate the map.
- **Filter by Module/Topic: Checkboxes** or **dropdowns** to highlight specific parts of the learning journey.
- **"Show Completed Paths" / "Show Remaining Paths" Toggles:** To filter what's visible on the map.
- **Clickable Nodes/Paths:** Interacting directly with the visual map (clicking on a task node) might open a pop-up with more details about that specific element.

Evaluation & Assessment Tools

1. Submission Review Interface

This is the primary place where evaluators (like mentors or instructors) go to review submitted work.

- **Student Selector/Search:** A **dropdown menu** or **search bar** to find a specific student's submission.
- **Assignment Selector:** A **dropdown menu** to choose which assignment's submissions to review.
- **Status Filters: Checkboxes** or **radio buttons** to filter submissions (e.g., "Pending Review," "Reviewed," "Graded").
- **Navigation Buttons:** "Previous Submission" and "Next Submission" buttons to move between different student submissions.
- **File Viewer/Player:** While not an input field, this is where the submitted document, code, video, or presentation would be displayed.
- **"Download Submission" Button:** To download the original submitted file.

2. Rubric-based Scoring

This allows for structured evaluation using customizable rubrics.

- **Rubric Name:** A **text box** for the rubric's title (e.g., "Research Paper Rubric").
- **Criteria Fields:** Multiple sets of fields for each evaluation criterion:
  - **Criterion Name:** A **text box** (e.g., "Content Accuracy," "Grammar and Punctuation").
  - **Description:** A **text area** to explain what's being evaluated.
  - **Rating Levels:** For each level (e.g., "Excellent," "Good," "Needs Improvement"):
    - **Level Name:** A **text box**.
    - **Points/Score:** A **number input field**.
    - **Description:** A **text area** explaining what constitutes that level of performance.
  - An **"Add New Criterion"** or **"Add New Level" button**.
- **Total Points Field:** A **number input field** to set the maximum possible score for the rubric.
- **"Save Rubric" Button:** To save the rubric for future use.

- **Input Fields (for using a rubric to score a submission):**
  - **Rubric Selector:** A **dropdown menu** to select which pre-made rubric to use.

- **Rating Selectors:** For each criterion, **radio buttons** or **dropdowns** to select the appropriate performance level, automatically calculating points.
- **Comments per Criterion:** A **text area** next to each criterion for specific feedback.
- **Override Score Field:** A **number input field** to manually adjust the final score if needed.

3. Multi-format Support

The system can evaluate various types of submissions.

- **"Upload File" Button:** When a student submits, a standard **file upload button** that accepts various file types (e.g., .doc, .pdf, .mp4, .zip). The system then uses its internal viewers/players.
- **Code Editor/Highlighter:** For code submissions, the interface would have a specialized area displaying code with line numbers and syntax highlighting.
- **Annotation Tools:** For documents or presentations, perhaps **drawing tools**, **text boxes for comments**, or **highlighting tools** that act as input fields to mark up the submission.
- **Video/Audio Controls:** Play, pause, scrub, volume controls for multimedia submissions.

4. Peer Review Management

Facilitates students evaluating each other's work.

- **Assignment Selector:** A **dropdown menu** to choose which assignment will have peer review.
- **"Enable Peer Review" Toggle:** An **on/off switch**.
- **Number of Reviews per Student:** A **number input field**.
- **Anonymity Settings: Radio buttons** (e.g., "Anonymous," "Show Reviewer Name").
- **Review Due Date:** A **date picker**.
- **"Assign Peers" Button:** To trigger the system to randomly assign reviewers, or possibly a manual **student selector** to assign them individually.

- **Input Fields (for students doing a peer review):**
  - **"Review Submission" Button:** A button next to an assigned peer's submission.
  - **Rubric-based Scoring:** (As described in point 2) for evaluating the peer's work.
  - **Overall Feedback:** A **text area** for general comments.
  - **"Submit Review" Button:** To finalize and submit their peer evaluation.

5. Automated Testing

Integration with code testing platforms for automated scoring of code.

- **"Attach Test Cases" Button:** A **file upload button** to upload test files
- **Expected Output Field:** A **text area** to define what the correct output should look like for specific inputs.
- **Test Script Editor:** A **text area** or dedicated code editor to write or paste automated test scripts.
- **"Run Tests" Button:** To test the student's code against predefined criteria.
- **Weightage Field:** A **number input field** to assign points for passing each test case.

- **Input Fields (for students):**
  - **"Run Tests" Button:** On their submission interface, to self-check their code against basic tests before final submission. (The results would be displayed, not input).

6. Feedback Management

Providing rich text feedback with multimedia support.

- **Overall Feedback Text Area:** A **rich text editor** (like a mini-word processor) where mentors can type detailed feedback, use bold/italics, create lists, etc.
- **"Insert Image" Button:** To upload and embed images into feedback.
- **"Record Audio/Video Feedback" Button:** To use a microphone/webcam to record a short audio or video message as feedback.
- **"Attach File" Button:** To upload and attach a separate file
- **"Send Feedback" / "Save Feedback" Button:** To deliver the feedback to the student or save it as a draft.

7. Grade Book

Comprehensive grade management and tracking.

- **Score Input Field:** A **number input field** next to each student's name for each assignment to enter their score. (Often auto-populated from rubric-based scoring).
- **Override Grade Checkbox:** A **checkbox** to allow manual entry of a final grade, overriding calculated scores.
- **Weightage Settings: Number input fields** next to each assignment category (e.g., "Quizzes: 20%," "Projects: 50%") to define how much each contributes to the final grade.
- **"Export Grades" Button:** To download the grade book as a CSV or Excel file.
- **"Calculate Final Grade" Button:** To manually trigger the calculation based on set weightages.
- **"Publish Grades" Button:** To make grades visible to students.

8. Performance Benchmarking

Comparing submissions against benchmarks.

- **Benchmark Selector:** A **dropdown menu** to select which benchmark to use for comparison (e.g., "Class Average," "Top 10%," "Expert Submission").
- **Submission Selector:** A **dropdown menu** to choose the student submission you want to compare.
- **Comparison Metrics: Checkboxes** to select which specific aspects of performance to compare (e.g., "Score," "Completion Time," "Adherence to Rubric Criteria").
- **"Run Benchmark" Button:** To display the comparison report.

9. Plagiarism Detection

Integrated plagiarism checking tools.

- **"Enable Plagiarism Check" Toggle/Checkbox:** A switch during assignment creation to automatically scan submissions.
- **"Run Plagiarism Check" Button:** A button on the submission review interface to manually run a check.
- **Exclusion Settings: Checkboxes** or **text fields** to exclude certain sources (e.g., "Exclude previous student submissions," "Exclude common phrases").
- **Similarity Threshold:** A **slider** or **number input field** to set the percentage above which a submission is flagged for review.

Communication & Mentoring Tools

1. Real-time Messaging

This feature lets you send instant messages to individual students or groups. Think of it like a chat app!

- **How it Works:** You'll see a chat window where you can type your messages.
    - **Text Input Box:** This is where you'll type your message. (e.g., a field labeled "Type your message here...")
    - **Recipient Selector (Dropdown/Search):** If starting a new chat, you might select a student or group from a list. (e.g., a dropdown to choose "Individual" or "Group," and then a search bar to find names).
    - **Send Button:** After typing, you'll click this to send your message.

2. Video Conferencing

Connect with students one-on-one through integrated video calls. It's like having a face-to-face conversation from anywhere!

- **How it Works:** When you initiate or join a video call, your camera and microphone will be used.
    - **"Start Video Call" Button:** You'd click this to begin a new video session with a selected student.
    - **"Join Call" Button/Link:** If someone invites you, you'd click a button or a link to enter the call.
    - **Microphone Toggle:** An icon you can click to mute or unmute your microphone.
    - **Camera Toggle:** An icon you can click to turn your camera on or off.

3. Screen Sharing

Share your computer screen to provide visual guidance, perfect for explaining technical concepts or demonstrating software.

- **How it Works:** During a video call, you can choose to share your entire screen or a specific application window.
    - **"Share Screen" Button:** Click this within a video call to initiate screen sharing.
    - **Screen/Window Selection Dialog:** A pop-up will appear, asking you to choose which screen or application you want to share. This isn't a traditional input field but a selection interface.
    - **"Stop Sharing" Button:** Once you're done, you'll click this to stop sharing your screen.

4. Virtual Whiteboard

A collaborative digital whiteboard for explaining ideas, drawing diagrams, and brainstorming together in real-time.

- **How it Works:** Both you and your student can draw, write, and add shapes on a shared digital canvas.
    - **Drawing Tools (Icons):** Buttons to select different drawing tools like pen, eraser, shapes, text box.
    - **Color Picker:** A palette to choose colors for your drawings.
    - **Text Input Field (on canvas):** When the text tool is selected, you'd click on the whiteboard and a text box would appear for you to type.

5. Discussion Forums

Create and moderate dedicated discussion spaces for students to ask questions, share insights, and interact with each other.

- **How it Works:** You can create new forum topics, post messages, and reply to others.
    - **"Create New Topic" Button:** Click this to start a new discussion thread.
    - **Topic Title Field:** A text box to enter the title of your discussion topic.
    - **Body/Content Text Area:** A larger text box where you'll type the main content of your post.
    - **"Post Reply" Button:** Click this when replying to an existing post.
    - **"Submit" Button:** To post your new topic or reply.

6. Office Hours

Set and manage virtual office hours for students to book consultations with you.

- **How it Works:** You define your availability, and students can see and book time slots.
    - **Date Picker:** To select the specific dates for your office hours.
    - **Time Slot Fields:** Input boxes or dropdowns to specify start and end times for each available slot. (e.g., "From: [Dropdown] To: [Dropdown]")
    - **"Add Slot" Button:** To add more available time slots.
    - **"Save Availability" Button:** To save your set office hours.
    - **"Book" Button (for students):** Students would click this on an available slot to reserve it.

7. Group Sessions

Conduct group mentoring sessions and workshops with multiple students.

- **How it Works:** Similar to one-on-one video conferencing, but designed for multiple participants. You'd schedule these sessions.
    - **"Schedule Group Session" Button:** To create a new group session.
    - **Session Title Field:** To name your group session.
    - **Date/Time Pickers:** To set the date and time of the session.
    - **Attendee Selector:** A field where you can search for and add students to invite to the session.
    - **"Create Session" Button:** To finalize the scheduling.

8. Recorded Sessions

Record mentoring sessions for students to review later. This is great for revision or for those who couldn't attend live.

- **How it Works:** During a live session, you'll have an option to start and stop recording. The recorded video will then be saved and accessible.
    - **"Start Recording" Button:** Click this during a live session to begin recording.
    - **"Stop Recording" Button:** Click this to end the recording.
    - **"View Recordings" Button/Link:** To access a list of past recorded sessions. (This would lead to a separate page with playable video files).

9. Chat History

Searchable archives of all your past chats and conversations, so you can easily refer back to previous discussions.

- **How it Works:** All your messages are saved and can be searched.
    - **Search Bar:** A text input field where you can type keywords to find specific messages or conversations. (e.g., "Search chat history...")
    - **Date Range Selector (Optional):** Input fields or dropdowns to filter chat history by a specific date range. (e.g., "From: [Date Picker] To: [Date Picker]")
    - **Filter Options (Dropdown):** To filter by student, group, or keyword.
    - **"Search" Button:** To execute your search query.

Content Creation & Course Management

1. Learning Content Creation

This allows you to create extra learning materials like text explanations, images, and embedded resources. Think of it as your digital canvas for teaching!

- **How it Works:** You'll use an editor to type, format, and add elements to your content.
    - **Content Title Field:** A text box for giving your learning material a clear name. (e.g., "Title: [Text input]")
    - **Rich Text Editor (Main Area):** This large area lets you type your content, add bold/italic text, lists, and even embed images or videos. It usually has formatting buttons above it (like a word processor).
    - **"Save Draft" Button:** To save your work as you go.
    - **"Publish" Button:** To make your content visible to students.

2. Video Content Tools

Record and edit your own instructional videos directly within the platform, making it easy to create dynamic lessons.

- **How it Works:** You can use your computer's camera and microphone to record, and then trim or add simple edits.
    - **"Start Recording" Button:** To begin capturing your video.
    - **"Stop Recording" Button:** To end the recording.
    - **Video Preview Window:** To watch your recorded video.
    - **Trim Handles (on a timeline):** Sliders or draggable points to cut unwanted parts from the beginning or end of your video.
    - **Video Title Field:** To name your video.
    - **"Save Video" Button:** To store your edited video.

3. Assignment Templates

Create reusable templates for assignments and projects, saving you time when setting up new tasks.

- **How it Works:** You design the structure and common instructions for an assignment once, and then reuse it for different classes or topics.
    - **Template Name Field:** To give your template a descriptive name.
    - **Assignment Description/Instructions (Rich Text Editor):** A large text area to write the general instructions for the assignment.
    - **Placeholder Fields:** You might add specific placeholders (e.g., [Due Date], [Specific Topic]) that you can fill in when using the template later. These aren't interactive input fields during creation but act as markers.
    - **"Save Template" Button:** To store your new template.

4. Resource Library

Build and organize your personal collection of resources specifically for mentoring, like articles, external links, or supplementary files.

- **How it Works:** You can upload files, link to external websites, and categorize your resources.
- **Input Fields:**
    - **"Add New Resource" Button:** To start adding a resource.
    - **Resource Type Selector (Dropdown):** To choose if you're adding a "File Upload," "Web Link," or "Text Note."
    - **File Upload Button:** A button that opens your computer's file explorer to select a file.
    - **URL Input Field:** For pasting web links. (e.g., "URL: [Text input]")
    - **Resource Title Field:** To give your resource a name.
    - **Category Selector (Dropdown/Text input):** To assign your resource to a category you create.
    - **"Save Resource" Button:** To add it to your library.

---

5. Quiz Creation

Design quizzes and assessments to evaluate student understanding and progress.

- **How it Works:** You'll create questions, set answer types, and define correct answers.
- **Input Fields:**
    - **"Create New Quiz" Button:** To start building a quiz.
    - **Quiz Title Field:** To name your quiz.
    - **"Add Question" Button:** To add a new question to the quiz.
    - **Question Type Selector (Dropdown):** To choose question types (e.g., "Multiple Choice," "True/False," "Short Answer").
    - **Question Text Area:** Where you type the actual question.
    - **Answer Option Fields (for MCQs):** Text boxes for each possible answer.
    - **Correct Answer Checkbox/Radio Button:** To mark the correct answer(s).
    - **Point Value Field:** A number input to set how many points the question is worth.
    - **"Save Quiz" Button:** To finalize and save your quiz.

---

6. Interactive Content

Create engaging tutorials and guided exercises that students can actively participate in.

- **How it Works:** This often involves a specialized editor that lets you add interactive elements like drag-and-drop, fill-in-the-blanks, or clickable hotspots.
- **Input Fields (Example for a "Fill-in-the-Blank"):**
    - **Content Text Area:** You'd type your sentence, and use a special button or syntax to create a blank.
    - **"Insert Blank" Button:** To add a fill-in-the-blank spot.
    - **Correct Answer Field (for blanks):** A text box that appears after creating a blank, where you type the correct answer.
    - **"Save Interactive Content" Button:** To save your creation.

---

7. Content Sharing

Easily share your created content with other mentors and students.

- **How it Works:** You'll select the content you want to share and then choose who should have access.
- **Input Fields:**
    - **Content Selection (Checkbox/List):** A list of your content where you can check boxes next to items to share.
    - **Share With Selector (Dropdown/Search):** To select specific students, groups, or other mentors you want to share with.
    - **Permissions Selector (Dropdown):** To choose if they can "View Only," "Edit," etc. (e.g., "Permissions: [Dropdown]")
    - **"Share" Button:** To finalize the sharing process.

---

8. Version Control

Manage different versions of your content and materials, allowing you to track changes and revert to older versions if needed.

- **How it Works:** The system automatically saves versions of your content as you make changes. You can then view the history and restore previous versions.
- **Input Fields (More about display and action buttons):**
    - **"Version History" Button/Link:** This would open a list of all saved versions of a particular piece of content.
    - **Version List (Display):** Shows different versions with details like "Date Modified," "Modified By," and potentially a "Notes" field.

    - **"Restore" Button:** Next to each old version, a button to revert your content back to that specific version.
    - **"Compare Versions" Button (Optional):** To visually see the differences between two versions.

Project and team management

1. Project Assignment

This feature lets you assign and manage real-world projects, making sure everyone knows what they need to do.

- **How it Works:** You'll create a new project and then specify its details, including who is responsible for it.
- **Input Fields:**
    - **Project Title Field:**
    - **Project Description Text Area:**
    - **Assignee Selector (Dropdown/Search):** A dropdown or search bar to select the mentor or team responsible for the project.
    - **Start Date Picker:** A calendar input to choose when the project officially begins.
    - **End Date Picker:** A calendar input to select the project's expected completion date.
    - **"Create Project" Button:** To save and initiate the new project.

---

2. Team Formation

Create and manage project teams, bringing together the right people for each task.

- **How it Works:** You can build a new team from scratch or modify an existing one, adding or removing members as needed.
- **Input Fields:**
    - **Team Name Field:**
    - **Member Selector (Search/Multi-select List):** A search bar or list where you can find and select individual students or mentors to add to the team.
    - **"Add Member" Button:** To confirm adding selected individuals to the team.
    - **"Remove Member" Button (next to each member):** An option to remove someone from the team.
    - **"Save Team" Button:** To finalize the team's composition.

---

3. Project Timeline

Set project milestones and track progress visually, helping you stay on schedule and identify potential delays.

- **How it Works:** You define key checkpoints (milestones) within the project's duration.
- **Input Fields:**
    - **"Add Milestone" Button:** To create a new milestone within a project's timeline.
    - **Milestone Name Field:** A text box for the name of the milestone (e.g., "Phase 1 Completion," "Final Review").
    - **Milestone Due Date Picker:** A calendar input to set the target date for completing this milestone.
    - **Progress Status Selector (Dropdown):** A dropdown to update the status of a milestone (e.g., "Not Started," "In Progress," "Completed").
    - **"Save Milestone" Button:** To add the milestone to the project timeline.

---

4. Resource Allocation

Manage project resources and requirements, ensuring that teams have everything they need to succeed.

- **How it Works:** You list the resources required for a project and assign them.
- **Input Fields:**
    - **"Add Resource" Button:** To add a new resource requirement for the project.
    - **Resource Name Field:** A text box for the name of the resource (e.g., "Software License," "Meeting Room," "Specific Hardware").
    - **Quantity Field (Number Input):** A numerical input for how many of that resource are needed.
    - **Assigned To Selector (Dropdown/Search):** To specify which team or individual the resource is allocated to.
    - **Availability Status Selector (Dropdown):** To track if the resource is "Available," "In Use," or "Requested."
    - **"Save Resource" Button:** To log the resource requirement.

---

5. Client Integration

Interface with real client projects and requirements, providing a bridge between academic projects and industry needs.

- **How it Works:** This feature allows you to link project details to external client information, often involving secure data exchange.
- **Input Fields:**
    - **"Link Client" Button:** To connect a project to a specific client.
    - **Client Name/ID Field:** A text box or dropdown to select an existing client or enter a new client's identifier.
    - **Client Project URL (Optional):** A text box to paste a link to the client's external project management system or documentation.
    - **API Key/Authentication Token Field (for secure connections):** A sensitive text input field if direct system-to-system integration is supported and requires credentials.
    - **"Save Integration" Button:** To establish the link.

---

6. Agile/Scrum Tools

Implement Agile methodology for project management, facilitating flexible and iterative development cycles.

- **How it Works:** You'll manage tasks in "sprints" and track their progress through various stages (e.g., "To Do," "In Progress," "Done").
- **Input Fields:**
    - **"Create New Sprint" Button:** To start a new iteration of work.
    - **Sprint Name Field:** To name the sprint (e.g., "Sprint 1," "Backend Development").
    - **Sprint Duration Selector (Dropdown/Number Input):** To define how long the sprint will last (e.g., "2 Weeks," "1 Month").
    - **Task/Story Name Field:** A text box to create a new task or user story within a sprint.
    - **Status Selector (Dropdown):** For each task, a dropdown to move it through workflow stages (e.g., "To Do," "In Progress," "Review," "Done").
    - **"Assign Task" Selector (Dropdown/Search):** To assign a task to a specific team member.
    - **"Save Sprint/Task" Button:** To commit changes.

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7. Code Review

Integrated code review and collaboration tools help teams improve code quality through peer feedback.

- **How it Works:** Developers submit their code for review, and others can comment, suggest changes, and approve.
- **Input Fields:**
    - **"Submit for Review" Button:** To mark a section of code or a file as ready for review.
    - **Reviewer Selector (Dropdown/Search):** To choose who should review the code.
    - **Comment Text Area (on code lines):** A text box that appears directly on the code viewer, allowing reviewers to type comments linked to specific lines of code.
    - **"Approve" / "Request Changes" Buttons:** Options for the reviewer to complete their feedback.
    - **"Reply to Comment" Text Area:** To respond to specific feedback.

---

8. Project Documentation

Manage project documentation and deliverables, keeping all important files organized and accessible.

- **How it Works:** You can upload, link, and organize all project-related documents.
- **Input Fields:**
    - **"Upload Document" Button:** To open a file browser and select a document to upload.
    - **Document Title Field:** A text box to give the uploaded document a clear title.
    - **Document Type Selector (Dropdown):** To categorize the document (e.g., "Requirements," "Design," "Meeting Minutes," "Final Report").
    - **Version Number Field (Optional):** A text or number input if you manually track versions.
    - **"Save Document" Button:** To add the document to the project's documentation library.
    - **Search Bar:** To find specific documents within the project's files.

Scheduling & Time Management

1. Integrated Calendar

This feature gives you a full calendar view to manage all your events, with the added benefit of syncing with external calendars you already use.

- **How it Works:** You'll see your schedule laid out, and you can add new events or view existing ones.
- **Input Fields:**
    - **"Add Event" Button:** Click this to create a new entry on your calendar.
    - **Event Title Field:** A text box to name your event (e.g., "Mentor Session with John," "Team Meeting").
    - **Start Date/Time Pickers:** Calendar and clock inputs to choose when the event begins.
    - **End Date/Time Pickers:** Calendar and clock inputs to choose when the event ends.
    - **Description Text Area:** A larger text box for adding details about the event.
    - **Location Field:** A text box for specifying where the event will take place (physical or virtual link).
    - **"Sync External Calendar" Button:** A button that, when clicked, might prompt you to select an external calendar service (e.g., Google Calendar) and authorize the connection.
    - **"Save Event" Button:** To add the event to your calendar.

---

2. Appointment Scheduling

Allow students to book mentoring sessions directly with you, simplifying the booking process for everyone.

- **How it Works:** You'll set your available slots, and students will see these slots and can book them.
- **Input Fields (for Mentors to set availability, similar to "Office Hours" but specific to appointments):**
    - **"Set Availability" Button:** To open your availability settings.
    - **Date Range Pickers:** To define the period for which you're setting availability.
    - **Daily Time Slot Fields:** Input fields (e.g., dropdowns or sliders) to specify your start and end times for each day of the week.

    - **Buffer Time Field (Number Input):** To add a small break between appointments (e.g., "15 minutes").
    - **"Save Availability" Button:** To update your bookable slots.
- **Input Fields (for Students to book):**
    - **Available Slot Selection (Clickable Blocks):** Students click on an available time slot displayed on your calendar.
    - **Purpose of Meeting Text Area:** A text box for students to briefly describe why they are booking the session.
    - **"Confirm Booking" Button:** To finalize their appointment.

---

3. Time Blocking

Block out specific periods in your calendar for different activities or to focus on particular tasks without interruptions.

- **How it Works:** You'll create dedicated blocks of time for focused work, just like a regular calendar event but specifically for personal productivity.
- **Input Fields:**
    - **"Block Time" Button:** To add a new time block.
    - **Block Title Field:** To name the activity or focus area (e.g., "Deep Work," "Project X Development").
    - **Start Date/Time Pickers:** To set when the blocked time begins.
    - **End Date/Time Pickers:** To set when the blocked time ends.
    - **Category/Tag Selector (Dropdown/Text Input):** To categorize the type of work being done (e.g., "Admin," "Coding," "Research").
    - **"Save Block" Button:** To add the time block to your schedule.

---

4. Recurring Events

Set up mentoring sessions, check-ins, or meetings that happen repeatedly, saving you from creating them individually each time.

- **How it Works:** When creating an event, you specify if and how it repeats.
- **Input Fields (added to event creation form):**
    - **"Repeats" Checkbox:** A checkbox to enable recurrence options.
    - **Recurrence Pattern Selector (Dropdown):** Choose how often it repeats (e.g., "Daily," "Weekly," "Monthly," "Custom").
    - **Frequency Field (Number Input):** If "Weekly," then "Every [Number] week(s)."

    - **Day of Week/Month Selector (Checkboxes/Dropdown):** If "Weekly," select "Mon, Wed, Fri." If "Monthly," select "15th of month" or "First Monday."
    - **End Recurrence Selector (Dropdown):** Choose when the recurrence stops (e.g., "Never," "After X occurrences," "On date: [Date Picker]").
    - **"Save Event with Recurrence" Button:** To save the recurring series.

---

5. Time Tracking

Track the time you spend on different mentoring activities, helping you understand your time allocation and productivity.

- **How it Works:** You can start a timer for an activity or manually log time spent.
- **Input Fields:**
    - **"Start Timer" Button:** Click this to begin tracking time for a new activity.
    - **Activity Name Field:** A text box to describe what you're working on (e.g., "Mentoring Session with Sarah," "Grading Assignments").
    - **"Stop Timer" Button:** To end the current time tracking.
    - **"Log Time Manually" Button:** To enter time that wasn't tracked by the timer.
    - **Manual Duration Fields (Number Input + Dropdown):** Enter "X hours" and "Y minutes."
    - **Date Picker (for manual log):** To select the date of the activity.
    - **Associated Project/Student Selector (Dropdown/Search):** To link the time spent to a specific project or student.
    - **"Save Log" Button:** To save the time entry.

---

6. Availability Management

Set complex availability patterns and preferences, allowing for detailed control over when you are open for bookings or meetings.

- **How it Works:** This is an advanced version of "Appointment Scheduling," letting you define intricate rules for your availability.
- **Input Fields (more detailed than simple time slots):**
    - **Default Working Hours (Start/End Time Pickers):** Set your general availability for weekdays.
    - **Specific Day Overrides (Date Picker + Time Slot Fields):** Add or remove availability for specific dates (e.g., "I'm off on July 4th," or "I'm only available from 1 PM-3 PM on Tuesdays").

    - **Recurring Exceptions (e.g., "Every Friday afternoon I'm unavailable"):** Similar to recurring events but for unavailability.
    - **Meeting Type Preferences (Checkboxes/Dropdown):** Specify if you're available for "Video Calls Only" or "In-person Meetings."
    - **Minimum Notice Period (Number Input + Dropdown):** How much advance notice you require for a booking (e.g., "24 Hours").
    - **"Save Availability Settings" Button:** To apply all your complex rules.

---

7. Meeting Management

Organize and manage meetings with students and colleagues efficiently.

- **How it Works:** You can create meetings, invite attendees, and share meeting agendas.
- **Input Fields (similar to "Integrated Calendar" but with meeting-specific additions):**
    - **"Schedule New Meeting" Button:** To create a new meeting.
    - **Meeting Title Field:** To name the meeting.
    - **Attendees Selector (Search/Multi-select List):** Find and invite students or colleagues to the meeting.
    - **Meeting Agenda Text Area:** A dedicated space to outline topics for discussion.
    - **Meeting Type Selector (Dropdown):** Choose between "Video Call," "In-person," or "Phone Call."
    - **Video Conference Link Field (auto-generated or manual):** If it's a video call, this might be automatically generated or you can paste a link.
    - **"Send Invitations" Button:** To send calendar invites to attendees.

---

8. Reminder System

Receive automated reminders for appointments and deadlines, ensuring you never miss an important event.

- **How it Works:** You'll set preferences for how and when you want to receive reminders.
- **Input Fields (found in user settings or when creating an event):**
    - **"Enable Reminders" Checkbox:** To turn reminders on or off.
    - **Reminder Method Selector (Checkboxes):** Choose how you want to be reminded (e.g., "Email," "In-App Notification," "SMS").
    - **Reminder Timing Selector (Number Input + Dropdown):** Specify when you want the reminder (e.g., "15 minutes before," "1 hour before," "1 day before").
    - **"Add Another Reminder" Button:** To set multiple reminders for the same event.

    - **Default Reminder Settings:** In your user profile, you might set your preferred default reminder times for all new events.
    - **"Save Reminder Settings" Button:** To apply your preferences.

Reporting & Performance Management

1. Student Performance Reports

Generate comprehensive reports to evaluate student progress and understanding.

- **How it Works:** You'll select the student(s) and specific criteria you want to report on.
- **Input Fields:**
    - **Student Selector (Dropdown/Search):** A dropdown or search bar to select one or more specific students for the report.
    - **Date Range Selector (Date Pickers):** Choose the start and end dates for the performance data you want to analyze (e.g., "From: [Date] To: [Date]").
    - **Report Type Selector (Checkboxes/Dropdown):** Select specific areas to include, such as "Quiz Scores," "Assignment Grades," "Participation," "Time Spent on Platform."
    - **"Generate Report" Button:** Click this to create the report.

---

2. Mentor Effectiveness Reports

Access reports for self-assessment and tracking your own performance as a mentor.

- **How it Works:** These reports focus on your activities and the outcomes of your mentoring efforts.
- **Input Fields:**
    - **Date Range Selector (Date Pickers):** Define the period for which you want to review your effectiveness.
    - **Metrics to Include (Checkboxes):** Select relevant metrics like "Number of Sessions Conducted," "Student Feedback Scores," "Average Session Duration," "Number of Students Mentored."
    - **"Generate Report" Button:** To view your effectiveness report.

---

3. Batch Performance Analysis

Analyze the performance trends across groups (batches) of students. This is great for identifying overall strengths and weaknesses within a cohort.

- **How it Works:** You'll select the student batch(es) and the performance indicators you're interested in.

- **Input Fields:**
    - **Batch Selector (Dropdown/Multi-select List):** Choose one or more student batches (e.g., "Batch A - 2024," "All Engineering Students").
    - **Performance Metric Selector (Dropdown/Checkboxes):** Select specific metrics for comparison, such as "Average Quiz Score," "Completion Rates," "Average Grade."
    - **Comparison Period Selector (Date Pickers/Dropdown):** To compare performance over specific periods or against previous batches.
    - **"Run Analysis" Button:** To generate the comparative report.

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4. Custom Report Builder

Create personalized reports by selecting specific data points and parameters.

- **How it Works:** This is your tool to build a report exactly how you need it, by picking and choosing data.
- **Input Fields:**
    - **Report Name Field:** A text box to give your custom report a name.
    - **Data Fields Selector (Multi-select List/Search):** A comprehensive list of all available data points (e.g., "Student Name," "Session Date," "Mentor Feedback," "Quiz Score," "Assignment Status"). You'd click to add them to your report.
    - **Filters/Conditions (Dropdowns & Text Fields):** Add rules to narrow down data (e.g., "Where 'Quiz Score' is > [Number input]," "Where 'Student Status' is 'Active' [Dropdown]").
    - **Grouping Option (Dropdown):** Choose how to group data (e.g., "Group by Student," "Group by Mentor," "Group by Date").
    - **Sorting Option (Dropdown):** Choose how to sort the data (e.g., "Sort by Date Ascending," "Sort by Grade Descending").
    - **"Preview Report" Button:** To see a sample of your custom report.
    - **"Save Custom Report" Button:** To save your report template for future use.

---

5. Automated Reporting

Schedule regular report generation and delivery, so you get the insights you need without manually running reports every time.

- **How it Works:** You configure a report (either a standard one or a custom one) to be sent to you at a specific frequency.
- **Input Fields:**
    - **Report Selection (Dropdown):** Choose which report you want to automate (from standard or saved custom reports).

    - **Frequency Selector (Dropdown):** How often should the report be generated (e.g., "Daily," "Weekly," "Monthly," "Quarterly").
    - **Delivery Method Selector (Checkboxes):** How should the report be delivered (e.g., "Email," "In-App Notification," "Slack Channel").
    - **Recipient Email(s) Field:** A text box to enter email addresses for delivery.
    - **Start Date Picker:** When should the automated reporting begin.
    - **"Schedule Report" Button:** To set up the automation.

---

6. Data Export

Export data from your reports in multiple formats, making it easy to use the information in other tools or for offline analysis.

- **How it Works:** After generating a report, you'll have options to download it in different file types.
- **Input Fields (usually buttons):**
    - **"Export as PDF" Button:** To download the report as a PDF document.
    - **"Export as Excel (XLSX)" Button:** To download the data in a spreadsheet format for detailed analysis.
    - **"Export as CSV" Button:** To download a plain text file of the data, commonly used for importing into other databases or tools.
    - **"Select Columns for Export" (Optional Checkboxes):** For some exports, you might be able to choose specific columns to include.

---

7. Performance Dashboards

Access visual dashboards that provide a quick, at-a-glance overview of key performance metrics.

- **How it Works:** Dashboards are pre-designed visual summaries. While you don't build them with input fields, you can often customize what you see on them.
- **Input Fields (for customization):**
    - **Time Period Selector (Dropdown):** To view data for "Today," "Last 7 Days," "This Month," "Last Quarter," "Custom Range."
    - **Filter Options (Dropdowns/Checkboxes):** To filter the data displayed on the dashboard (e.g., "Filter by Batch," "Filter by Mentor," "Filter by Course").
    - **Widget Reordering/Toggle (Drag & Drop/Checkboxes):** To customize which visual elements (widgets) appear on your dashboard.
    - **"Apply Filters" Button:** To update the dashboard view.

---

8. Trend Analysis

Perform long-term trend analysis and gain deeper insights into performance patterns over extended periods.

- **How it Works:** This feature helps you see how performance changes over time, identifying improvements or areas needing attention. It often uses the same underlying data as reports but focuses on visualization over time.
- **Input Fields (similar to report building, but focused on time-series data):**
    - **Data Series Selector (Multi-select List):** Choose the specific metrics you want to plot on a trend line (e.g., "Average Quiz Score," "Student Engagement Hours").
    - **Time Granularity Selector (Dropdown):** Choose how the data points are grouped (e.g., "Daily," "Weekly," "Monthly," "Quarterly").
    - **Comparison Period Selector (Date Pickers):** Select a primary period and an optional comparison period to see changes.
    - **Breakdown By Selector (Dropdown):** To analyze trends for different segments (e.g., "Breakdown by Gender," "Breakdown by Mentor," "Breakdown by Course Type").
    - **"Generate Trend Chart" Button:** To visualize the data trends.

Professional Development Tools

1. Skill Development Tracking

Track your own skill development as a mentor, identifying areas of growth and mastery.

- **How it Works:** You'll log your skills, assess your proficiency, and update your progress over time.
- **Input Fields:**
    - **"Add Skill" Button:** To add a new skill you want to track (e.g., "Public Speaking," "Python Programming").
    - **Skill Name Field:** A text box to type the name of the skill.
    - **Proficiency Level Selector (Dropdown/Slider):** A dropdown (e.g., "Beginner," "Intermediate," "Advanced," "Expert") or a slider to rate your current proficiency.
    - **Date Achieved/Updated (Date Picker):** To log when you gained or updated a certain level of skill.
    - **Notes Text Area:** For any personal notes about your development in this skill.
    - **"Save Skill" Button:** To add or update the skill in your profile.

---

2. Training Module Access

Access a library of training and development content specifically designed for mentors.

- **How it Works:** You'll browse, select, and launch training modules, which might be videos, interactive courses, or documents.
- **Input Fields (mainly for searching and filtering):**
    - **Search Bar:** A text input field to search for specific training modules by keyword (e.g., "active listening," "feedback techniques").
    - **Category Filter (Dropdown/Checkboxes):** To narrow down modules by category (e.g., "Communication," "Technical Skills," "Soft Skills").
    - **Progress Tracking (Display/Action):** While not an input field, you might see "Resume Module" buttons or progress bars indicating your completion.
    - **"Enroll/Start Module" Button:** To begin a chosen training module.

---

3. Certification Management

Track and manage your professional certifications, ensuring all your credentials are up-to-date and organized.

- **How it Works:** You'll add details about your certifications, including expiration dates and supporting documents.
- **Input Fields:**
    - **"Add Certification" Button:** To add a new professional certification.
    - **Certification Name Field:** A text box for the name of the certification (e.g., "Certified Scrum Master," "Google Cloud Professional Architect").
    - **Issuing Organization Field:** A text box for the organization that issued the certification.
    - **Issue Date (Date Picker):** To select when the certification was obtained.
    - **Expiration Date (Date Picker):** To set when the certification expires.
    - **Certificate ID/Number Field (Optional):** A text box for any unique identifier on your certificate.
    - **Upload Certificate Button:** To upload a digital copy of your certificate (e.g., PDF, image).
    - **"Save Certification" Button:** To record your certification details.

---

4. Best Practices Library

Access a curated collection of mentoring best practices, guides, and resources.

- **How it Works:** This is a searchable and browsable library of helpful information.
- **Input Fields (for finding information):**
    - **Search Bar:** A text input field to search for best practices by topic (e.g., "effective questioning," "setting boundaries").
    - **Category Filter (Dropdown/Checkboxes):** To filter resources by relevant categories (e.g., "Initial Sessions," "Conflict Resolution," "Career Guidance").
    - **Rating/Favorite Button:** While not direct input, you might click a star or heart icon to rate or save a best practice for easy access later.

---

5. Peer Learning

Collaborate and learn from other mentors through shared experiences and direct interaction.

- **How it Works:** This could involve dedicated forums, group chats, or collaborative document spaces.
- **Input Fields:**
    - **"Join Peer Group" Button:** To join a specific group of mentors.
    - **"Start Discussion" Button:** To initiate a new discussion topic within a peer group.
    - **Discussion Title Field:** For the title of your discussion.
    - **Message/Post Text Area:** A text box to type your contributions to discussions.
    - **"Reply" Button:** To respond to another mentor's post.
    - **Mentor Search/Connect Feature:** A search bar to find and connect with specific mentors for one-on-one peer learning.

---

6. Webinar Integration

Attend and organize professional development webinars, expanding your knowledge and skills through live online sessions.

- **How it Works:** You can register for upcoming webinars or schedule your own.
- **Input Fields (for attending):**
    - **"Register for Webinar" Button:** To sign up for a scheduled webinar.
- **Input Fields (for organizing):**
    - **"Schedule New Webinar" Button:** To set up your own professional development webinar.

    - **Webinar Title Field:** To name your webinar.
    - **Date/Time Pickers:** To set the date and time of the webinar.
    - **Description Text Area:** To provide details about the webinar's content.
    - **Invitee Selector (Search/Multi-select):** To choose which mentors or community members to invite.
    - **Platform Selector (Dropdown):** If integrated with multiple platforms (e.g., Zoom, Google Meet), to choose which one to use.
    - **"Schedule Webinar" Button:** To finalize the webinar setup.

---

7. Knowledge Sharing

Share your experiences and insights with the broader mentor community, contributing to collective learning.

- **How it Works:** You can contribute articles, tips, or case studies to a shared knowledge base.
- **Input Fields:**
    - **"Share Knowledge" Button:** To start creating a new knowledge sharing entry.
    - **Title Field:** For the title of your shared insight or article.
    - **Content Editor (Rich Text Area):** A large text area where you type your experiences, insights, or best practices, with formatting options.
    - **Tags/Keywords Field:** A text input to add relevant tags to make your content discoverable (e.g., "onboarding," "motivation," "technical").
    - **"Submit for Review/Publish" Button:** To submit your contribution to the knowledge base.

---

8. Career Path Planning

Access professional development and career planning tools to map out your mentoring journey and future career goals.

- **How it Works:** You'll set goals, identify necessary skills, and track your progress towards career milestones.
- **Input Fields:**
    - **"Set Career Goal" Button:** To define a new career objective (e.g., "Lead Mentor," "Specialist in AI Mentoring").
    - **Goal Description Text Area:** To elaborate on your goal.
    - **Target Completion Date (Date Picker):** To set a deadline for your goal.

- **Required Skills List (Checkboxes/Text Input):** To list skills needed to achieve the goal, possibly linking to skill development tracking.
- **"Add Action Item" Button:** To add specific steps needed to reach your goal.
- **Action Item Description Field:** For the task you need to complete.
- **Action Item Due Date (Date Picker):** For the task's deadline.
- **"Save Plan" Button:** To save your career development plan.

🚀 **Integration & Automation Features**

1. Learning Platform Integration

What it does: Connects our system with other learning platforms (like Moodle, Canvas, or Blackboard) you might be using. This means you can share information between them, perhaps for tracking student progress or sharing course materials.

How it works (Input Fields):

- **Learning Platform Type:** (Dropdown menu: Moodle, Canvas, Blackboard, Other)
- **Platform URL/Address:** (Text field: e.g., https://your-school.moodle.com)
- **API Key/Credentials:** (Text field: *This is a secret code that gives our system permission to talk to your learning platform. You'll usually get this from your learning platform's settings.*)
- **Data to Sync:** (Checkboxes: Student Grades, Course Assignments, User Accounts, etc.)

---

2. Code Repository Integration

What it does: Connects our system with popular code storage services like GitHub, GitLab, and Bitbucket. This is super helpful for developers to manage their code and automate tasks like testing or deployment.

How it works (Input Fields):

- **Code Repository Service:** (Dropdown menu: GitHub, GitLab, Bitbucket)
- **Repository URL:** (Text field: e.g., https://github.com/your-username/your-project)
- **Access Token/Credentials:** (Text field: *A secret key to let our system access your code. You'll generate this in your code repository settings.*)
- **Events to Monitor:** (Checkboxes: Code Push, Pull Request, Issue Creation, etc.)

---

3. Communication Tool Integration

What it does: Links our system with communication apps like Slack or Microsoft Teams. This allows our system to send notifications or messages directly into your team chats, keeping everyone updated.

How it works (Input Fields):

- **Communication Tool:** (Dropdown menu: Slack, Microsoft Teams)
- **Channel Name/ID:** (Text field: e.g., #general-announcements or a specific channel ID)
- **Webhook URL:** (Text field: *A unique link provided by your communication tool for sending messages.*)
- **Notification Triggers:** (Checkboxes: Task Completed, New User Registered, Error Occurred, etc.)

---

4. Calendar Integration

What it does: Syncs our system with your calendar apps like Google Calendar or Outlook. This means events or deadlines from our system can show up directly in your calendar, and vice-versa.

How it works (Input Fields):

- **Calendar Service:** (Dropdown menu: Google Calendar, Outlook Calendar, Apple Calendar, Other)
- **Account Login/Authorization:** (Button: "Connect to Google/Outlook" - *This will open a pop-up asking for your permission to link your calendar.*)
- **Calendar to Sync:** (Dropdown menu: Your Main Calendar, Specific Project Calendar)
- **Sync Direction:** (Radio buttons: One-way (Our System -> Calendar), Two-way (Both ways))

---

5. Automated Workflows

What it does: Allows you to set up automatic "if this, then that" rules for routine tasks. For example, "If a new user signs up, then send them a welcome email."

How it works (Input Fields):

- **Trigger Event:** (Dropdown menu: New User Created, Task Status Changed, File Uploaded, etc.)
- **Conditions (Optional):** (Text field: e.g., If user's role is "Admin")
- **Action to Perform:** (Dropdown menu: Send Email, Create Task, Update Record, Send Notification, etc.)
- **Action Details:** (Text fields vary based on action: Email Subject, Task Due Date, Notification Message)

---

6. Custom Integrations

What it does: Provides flexibility to connect our system with almost any other tool or platform, even if it's not listed above. This is for more unique or specialized needs.

How it works (Input Fields):

- **Target System/Platform Name:** (Text field: e.g., Our Internal CRM)
- **Integration Method:** (Radio buttons: API, Webhook, Custom Script)
- **Connection Details:** (Text Area: *Provide technical details here, like API endpoints, data formats, or script code. This might require some technical knowledge.*)
- **Expected Data Format:** (Dropdown menu: JSON, XML, CSV, Other)

---

7. API Access

What it does: Gives you your own personal "key" (API access) to our system. This means you (or another system you use) can programmatically send commands to our system or pull information from it. It's like having a direct line of communication with our system's brain!

How it works (Input Fields):

- **Generate API Key:** (Button: "Generate New API Key" - *This will create a unique string of characters for you.*)
- **API Key Name (Optional):** (Text field: e.g., My Reporting Tool API Key)
- **Permissions:** (Checkboxes: Read Data, Write Data, Create Users, etc. - *Choose what actions this API key is allowed to perform.*)

---

8. Webhook Configuration

What it does: Allows our system to send real-time updates to other systems when something specific happens. Imagine our system yelling, "Hey! Something just changed!" to another system the moment it happens.

How it works (Input Fields):

- **Event to Trigger Webhook:** (Dropdown menu: New Item Created, Item Updated, Item Deleted, etc.)
- **Webhook URL:** (Text field: e.g., https://your-other-system.com/webhook-listener)
- **Secret Key (Optional):** (Text field: *A secret code to verify that the update is truly coming from our system.*)
- **Data Format:** (Dropdown menu: JSON, XML, Plain Text)

📱 Mobile & Accessibility Features

1. Mobile-Optimized Interface

What it does: This means our system looks and works perfectly on your phone or tablet, not just on a computer screen. You get the **full experience and all the features**, no matter what device you're using. It's like the system automatically reshapes itself to fit your screen!

How it works (No direct input fields, it's automatic!):

- **Simply access our system from your mobile device.** The interface will automatically adjust for optimal viewing and interaction.

---

2. Push Notifications

What it does: Sends you instant alerts directly to your phone or tablet, even when you're not actively using our system. Think of them as quick heads-up for important updates, new messages, or upcoming deadlines.

How it works (Input Fields):

- **Enable Push Notifications:** (Toggle switch: ON/OFF)
- **Receive Notifications For:** (Checkboxes: New Messages, Task Deadlines, System Updates, Mentions, etc.)
- **Notification Sound:** (Dropdown menu: Default, Custom Sound, None)
- **Do Not Disturb Hours:** (Time range picker: e.g., 10:00 PM - 7:00 AM)

---

3. Offline Capability

What it does: Lets you access and even work with certain parts of our system even when you don't have an internet connection. This is fantastic for when you're on the go or in an area with spotty Wi-Fi. Your work syncs up once you're back online!

How it works (Input Fields):

- **Enable Offline Access:** (Toggle switch: ON/OFF)
- **Download for Offline Use:** (Checkboxes: My Tasks, Saved Documents, Specific Courses/Content, etc.)
- **Storage Limit for Offline Data:** (Slider/Text field: e.g., 500 MB, 1 GB)
    - *Note: Our system will automatically manage what's available offline to save space.*

---

4. Voice Commands

What it does: Allows you to control our system using your voice! Instead of typing or clicking, you can simply speak commands to navigate, open features, or even create content. This is great for hands-free operation or for those who prefer voice interaction.

How it works (Input Fields):

- **Enable Voice Commands:** (Toggle switch: ON/OFF)
- **Voice Activation Phrase (Optional):** (Text field: e.g., "Hey System", "Listen Up")
- **Voice Input Language:** (Dropdown menu: English, Spanish, Hindi, etc.)
- **Microphone Access:** (Button: "Allow Microphone Access" - *You'll need to grant permission to your device.*)

---

5. Accessibility Compliance (WCAG 2.1) 👨🦯

What it does: This means our system is built to be usable by as many people as possible, including those with disabilities. **WCAG 2.1** is a set of guidelines that ensures our system is:

- **Perceivable:** Information and user interface components are presentable to users in ways they can perceive (e.g., text can be read by screen readers).
- **Operable:** User interface components and navigation are operable (e.g., keyboard navigation).
- **Understandable:** Information and the operation of user interface are understandable.
- **Robust:** Content can be interpreted reliably by a wide variety of user agents, including assistive technologies.

How it works (Input Fields for customization):

- **Screen Reader Optimization:** (Toggle switch: ON/OFF - *This optimizes the system for software that reads content aloud for visually impaired users.*)
- **High Contrast Mode:** (Toggle switch: ON/OFF - *Changes colors for better readability, especially for users with visual impairments.*)
- **Font Size Adjustment:** (Slider: Small, Medium, Large, Extra Large)
- **Reduced Motion:** (Toggle switch: ON/OFF - *Minimizes animations and flashy effects for users sensitive to motion.*)

---

6. Multi-language Support

What it does: Our system can be used in different languages. This is super helpful for international users, allowing them to interact with the system in their native tongue.

How it works (Input Field):

- **Interface Language:** (Dropdown menu: English, Spanish, Hindi, French, German, etc.)
    - *Note: Changing this will translate all menus, buttons, and system messages.*

---

7. Dark Mode

What it does: Offers a dark theme for the interface. This makes the screen less bright, which can be easier on your eyes, especially in low-light conditions or for extended screen time. It also often saves battery life on devices with OLED screens.

How it works (Input Field):

- **Theme:** (Radio buttons: Light Mode, Dark Mode, System Default - *Matches your device's system-wide theme setting.*)

---

8. Customizable Interface

What it does: Lets you personalize how the system looks and feels to match your preferences. You can arrange elements, choose colors, or decide what information is most prominent for you.

How it works (Input Fields):

- **Dashboard Layout:** (Drag-and-drop elements: Rearrange widgets, hide/show sections)
- **Accent Color:** (Color picker: Choose a highlight color for buttons and links)
- **Default View:** (Dropdown menu: Home Screen, My Tasks, Dashboard, etc. - *What you see first when you log in.*)
- **Show/Hide Specific Features:** (Checkboxes: Calendar, Notifications Panel, Quick Links)

Security & Data Privacy

1. Data Encryption

What it does: This is like putting your sensitive information (like mentor details or student grades) into a secret code. Even if someone were to get their hands on it, they wouldn't be able to understand it without the special "key" to decrypt it. We **encrypt all sensitive mentor and student data** at rest (when it's stored) and in transit (when it's moving).

How it works (No direct user input; it's automatic and fundamental):

- **Our system automatically encrypts all sensitive data.** You don't need to do anything to enable this; it's a core part of how we protect your information.

2. Secure Communication

What it does: Ensures that any conversations or data shared between you and our system, or between different users within the system, are completely private. This is achieved through **end-to-end encrypted communication channels**, meaning only the sender and the intended recipient can read the messages. It's like having a secure, private phone line.

How it works (No direct user input; it's automatic):

- All communication within and with the system, including messages and data transfers, is automatically secured.

---

3. Access Control

What it does: Determines who can see what and do what within the system. We provide **granular permissions for different mentor functionalities**, meaning we can set very specific rules. For example, a mentor might be able to view student progress but not edit their personal profiles, or an administrator might have full access to everything.

How it works (Input Fields typically for administrators/managers):

- **User Role/Group:** (Dropdown menu: Administrator, Mentor, Student, Observer)
- **Permissions for [Selected Role]:** (Checkboxes/Toggles for specific actions):
    - View Student Profiles (Allow/Deny)
    - Edit Course Content (Allow/Deny)
    - Manage User Accounts (Allow/Deny)
    - Access Financial Reports (Allow/Deny)
    - Send Announcements (Allow/Deny)
- **Specific User Override (Optional):** (Search bar to select a user)
    - Override Permissions for [Selected User] (Checkboxes/Toggles as above)

---

4. Session Security

What it does: Keeps your logged-in session safe. When you log in, you create a "session." **Secure session management with timeout controls** means we use methods to make sure your session isn't easily hijacked. Plus, it will automatically log you out after a period of inactivity to prevent unauthorized access if you leave your computer unattended.

How it works (Input Fields, typically for user preferences or admin settings):

- **Automatic Logout after Inactivity:** (Dropdown menu: 15 minutes, 30 minutes, 1 hour, 2 hours, Never - *Not recommended for security.*)
- **Require Re-authentication for Sensitive Actions:** (Toggle switch: ON/OFF - *e.g., asking for your password again before changing critical settings.*)
- **Remember Me (on this device):** (Checkbox: Yes/No - *This remembers your login for a certain period but is still subject to timeout controls.*)

---

5. Audit Logging

What it does: Keeps a detailed record of **comprehensive logging of mentor activities** (and other user activities). This is like a security camera that records who did what, when, and from where. It's crucial for tracking changes, troubleshooting issues, and investigating any suspicious activity.

How it works (Input Fields, typically for administrators or security personnel to view):

- **View Audit Logs:** (Button: "Access Activity Log")
- **Filter Logs By:**
    - User Name: (Text field)
    - Activity Type: (Dropdown: Login, Logout, Data Modified, File Uploaded, etc.)
    - Date Range: (Date picker: From/To)
- **Export Logs:** (Button: "Download Log Report")

---

6. Privacy Controls

What it does: Gives you direct control over your own personal information. You can **manage personal data visibility and sharing**, deciding what parts of your profile are public, who can see certain details, and if your data can be used for specific purposes (like research or marketing).

How it works (Input Fields, for individual users to manage their own privacy):

- **Profile Visibility:** (Radio buttons: Public, Only Connected Users, Private)
- **Data Sharing Preferences:** (Checkboxes):
    - Allow my data for anonymized research
    - Share my email for system updates/newsletters
- **Control Data Access for Specific Features:** (Toggles/Checkboxes):
    - Show my online status
    - Allow others to view my activity feed
- **Request Data Export:** (Button: "Download My Data")
- **Request Data Deletion:** (Button: "Delete My Account/Data" - *Subject to legal and retention policies.*)

---

7. GDPR Compliance

What it does: This means our system fully adheres to the **General Data Protection Regulation (GDPR)**, which is a very strict data privacy law originating in Europe. Even if you're not in Europe, compliance with GDPR demonstrates our commitment to high standards of data protection, user rights, and transparency regarding how your data is collected, processed, and stored.

How it works (No direct user input beyond accepting privacy policies; it's a legal and technical framework):

- **Review Privacy Policy:** (Link: "Read our GDPR-Compliant Privacy Policy")
- **Cookie Consent:** (Pop-up/Banner: "Accept Cookies" / "Manage Preferences")
    - *This ensures we get your permission before tracking your website activity.*

- **Data Subject Rights:** (Links/Forms for users to exercise their rights, e.g., "Request my data," "Correct my data," "Delete my data")

---

8. Secure File Sharing

What it does: Ensures that any files you upload, store, or share within our system are protected. This involves **encrypted file sharing and storage**, meaning your documents, images, and other files are securely stored and can only be accessed by authorized individuals.

How it works (Input Fields):

- **File Upload:** (Button: "Upload File")
- **Share With:** (Dropdown/Search field: Select Specific Users, Groups, or Public Link)
- **Access Permissions for Shared File:** (Radio buttons/Checkboxes):
    - View Only
    - Can Edit
    - Can Download
- **Password Protect Shared Link (Optional):** (Text field: Enter Password)
- **Expiration Date for Shared Link (Optional):** (Date picker)

Analytics & Intelligence Features

1. Student Success Prediction

What it does: Uses AI to **predict how likely a student is to succeed** based on various factors like their past performance, engagement, and activity within the system. This helps mentors identify students who might need extra support *before* they struggle.

How it works (Input Fields, typically for mentors/administrators to view and configure):

- **View Predictions For:** (Dropdown menu: All Students, My Mentees, Specific Group)
- **Prediction Threshold:** (Slider: e.g., Show students with < 60% predicted success rate)
- **Factors to Consider (for prediction model):** (Checkboxes: Assignment Grades, Login Frequency, Communication Activity, Course Completion Rates)
- **Generate Prediction Report:** (Button: "Run Prediction Report")

---

2. Mentoring Effectiveness Analysis

What it does: Helps mentors understand **how effective their mentoring techniques are**. It analyzes mentoring approaches and their outcomes, allowing mentors to see what's working well and what could be improved.

How it works (Input Fields, for mentors to view their performance):

- **View My Effectiveness Report:** (Button: "Generate Mentoring Effectiveness Report")
- **Filter Report By:**
    - Time Period: (Dropdown: Last Month, Last Quarter, All Time)
    - Student Group: (Dropdown: All Mentees, High-Performing Mentees, At-Risk Mentees)
- **Key Metrics to Analyze:** (Checkboxes: Student Engagement Scores, Student Achievement Gains, Mentee Feedback Scores)

---

3. Learning Path Optimization

What it does: Uses AI to **suggest the best possible learning paths for students**. Based on a student's strengths, weaknesses, learning style, and goals, the system can recommend specific resources, courses, or activities to help them learn most efficiently.

How it works (Input Fields, typically for students or mentors configuring for students):

- **Student Learning Goals:** (Text field: e.g., "Improve Algebra skills", "Prepare for SAT")
- **Preferred Learning Style:** (Radio buttons: Visual, Auditory, Kinesthetic, Reading/Writing)
- **Areas of Interest:** (Dropdown/Multi-select: Science, History, Coding, Arts)
- **Generate Optimized Path:** (Button: "Suggest My Learning Path")
- **Review/Adjust Path:** (Interactive display of suggested path with options to "Add Module," "Remove Activity")

---

4. Risk Assessment

What it does: Actively **identifies students who are at risk of dropping out or underperforming**. This is similar to student success prediction but specifically focuses on identifying potential problems early, allowing for timely intervention.

How it works (Input Fields, for mentors/administrators to monitor):

- **View At-Risk Students:** (Button: "Show At-Risk Students List")
- **Risk Indicators:** (Checkboxes: Low Attendance, Missed Deadlines, Declining Grades, Low Engagement)
- **Alert Threshold:** (Slider: e.g., Alert me if risk score > 70%)
- **Intervention Suggestions:** (Optional display of recommended actions for identified at-risk students)

---

5. Personalized Recommendations

What it does: Provides **AI-driven recommendations for mentoring approaches** tailored to individual students. Instead of generic advice, mentors receive suggestions on how best to interact with a specific student, what topics to focus on, or even what communication style might be most effective.

How it works (Input Fields, for mentors to get advice for specific mentees):

- **Select Mentee:** (Dropdown/Search field: Choose a specific student)
- **Get Mentoring Recommendations For:** (Button: "Generate Personalized Advice")
- **Focus Area for Recommendation:** (Dropdown: Academic Support, Motivational Guidance, Skill Development)

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6. Sentiment Analysis

What it does: Analyzes the tone and emotion in **student communication for engagement levels**. This can help understand how students are feeling about their learning, a course, or a mentoring relationship, by looking at their messages, forum posts, or feedback.

How it works (No direct input fields; the analysis happens behind the scenes):

- **View Sentiment Trends:** (Dashboard/Graph showing overall sentiment or sentiment per student/group)
- **Flag Negative/Positive Sentiment Alerts:** (Toggle switch: ON/OFF)
- **Keyword Analysis:** (Displays common positive/negative words used in student communications)

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7. Pattern Recognition

What it does: Identifies **successful mentoring patterns and strategies**. By analyzing data from many mentoring relationships, the system can spot recurring actions or approaches that lead to positive student outcomes, sharing these insights as best practices.

How it works (Input Fields, for administrators/mentoring program managers):

- **Identify Successful Patterns Based On:** (Checkboxes: High Student Retention, Significant Grade Improvement, Positive Student Feedback)
- **Generate Pattern Report:** (Button: "Analyze Successful Mentoring Patterns")
- **Filter Patterns By:** (Dropdown: Mentoring Style, Communication Frequency, Resource Usage)

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8. Adaptive Learning

What it does: Allows the system to **adapt mentoring strategies based on student responses** and performance in real-time. If a student is struggling with a concept, the system might suggest a different explanation or more practice. If they're excelling, it might recommend more challenging material.

How it works (Input Fields, generally an automatic process with some configuration):

- **Enable Adaptive Strategies:** (Toggle switch: ON/OFF)
- **Adaptation Level:** (Slider: Low, Medium, High - *Determines how aggressively the system adapts.*)
- **Feedback Loop Settings:** (Checkboxes: Adjust based on Quiz Scores, Assignment Submissions, Interaction Data)
- **Preferred Remediation Methods:** (Dropdown/Multi-select: Extra Exercises, Video Explanations, Peer Tutoring Suggestions)

Technical Implementation Stack

1. React.js 18+ with TypeScript

- **What it is:** Imagine building a complex LEGO set. **React.js** is like the ultimate instruction manual and a set of specialized tools that help us build the different "pieces" (like buttons, forms, or entire sections of a page) of our system very efficiently. It's designed to make our system super fast and responsive as you click around. The "18+" means we're using a very recent and powerful version.
- **What is TypeScript?** This is like having an extra helpful assistant for our LEGO builders. **TypeScript** is a special "flavor" of JavaScript (the programming language of the web) that helps our developers write code with fewer mistakes and makes it easier for multiple developers to work together on big projects. It acts like a spell-checker and grammar-checker for our code, catching errors *before* they become a problem.
- **Why it's important for you:** This combination means you get a **smooth, fast, and reliable experience**. Pages load quickly, interactions feel instant, and the system is less likely to have unexpected glitches.

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2. Redux Toolkit for State Management

- **What it is:** Think of our system as a giant online store. There's a lot of "information" that needs to be kept track of: what's in your shopping cart, whether you're logged in, what products you're currently viewing, etc. This "information" is called **"state." Redux Toolkit** is like a super-organized central control room or a universal clipboard that holds *all* this information consistently across our entire system. When something changes (like adding an item to your cart), Redux Toolkit makes sure every part of the system that needs to know about that change is updated immediately and correctly.
- **Why it's important for you:** This ensures a **consistent and predictable experience**. If you add an item to your cart, that item immediately shows up in your cart icon, even if you navigate to a different page. It prevents confusing situations where one part of the system doesn't "know" what another part is doing.

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3. Material-UI or Tailwind CSS for Components

- **What it is:** These are like pre-made, high-quality design kits for building the visual parts of our system.
    - **Material-UI:** This is a library of beautiful, pre-designed user interface (UI) elements (like buttons, navigation bars, cards, forms) that follow Google's "Material Design" guidelines. It's like having a set of polished, ready-to-use building blocks that look consistent and professional.
    - **Tailwind CSS:** Instead of pre-made blocks, Tailwind CSS provides a highly flexible set of "utility classes." Imagine it as a giant toolbox filled with individual styling tools (like "make text bold," "add small margin," "set background to blue"). Our developers use these tools directly in their code to build custom designs very quickly and efficiently.
- **Why it's important for you:** Both options are chosen for building a **clean, intuitive, and visually appealing interface**. Whether it's the consistent look of Material-UI or the tailored feel achieved with Tailwind, the goal is to make the system pleasant to look at and easy to navigate.

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4. React Router for Navigation

- **What it is:** When you click a link or a button in a website, you move from one "page" or "view" to another. **React Router** is the system that handles this smooth navigation within our single-page application. Instead of loading an entirely new page from scratch every time you click something, React Router cleverly swaps out just the necessary parts of the page, making the experience feel incredibly fast and seamless.
- **Why it's important for you:** This means **blazing-fast navigation**. You won't see annoying full-page reloads every time you click on a different section, making your interaction with the system feel much more fluid and modern, similar to using a desktop application.

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5. Socket.io client for Real-time Features

- **What it is:** Imagine a regular website where you have to constantly refresh the page to see new messages or updates. That's not very "real-time"! **Socket.io client** is a technology that creates a continuous, two-way, open communication channel between your device (the "client") and our system's servers. It's like having a constantly open phone line where information can flow instantly in both directions without you having to ask for it.
- **Why it's important for you:** This powers all the **real-time features** in our system. Think instant chat messages, live notifications, collaborative editing, or immediate updates to dashboards without needing to refresh your browser. It makes the system feel dynamic and truly live.

Backend Integration Explained

Backend integration is all about how the parts of an application you **see and interact with** (the frontend, like a website or mobile app) talk to the parts you **don't see** (the backend, which handles data, logic, and server-side operations). Think of it like a restaurant:

- **Frontend:** The dining area, the menu, and the waiters (what the customer sees and interacts with).
- **Backend:** The kitchen, the chef, the recipes, and the pantry (where all the behind-the-scenes work happens).

Just as a waiter takes your order to the kitchen, backend integration allows your app to send requests to the server and receive information back.

How Input Fields Work with Backend Integration

Input fields are where you, the user, provide information to the application. When you type something into an input field and click a button (like "Submit" or "Save"), that information often needs to be sent to the backend for processing, storage, or to trigger an action.

Let's look at how input fields relate to each point from your image:

1. RESTful API Consumption

- **What it is:** Imagine a waiter (your app) having a specific menu (API) to order different dishes (data) from the kitchen (backend). RESTful APIs are like a standardized way for your app to request and send data to the backend. It's a common way for web services to communicate.
- **How input fields work:**
    - **Search Bar:** You type a product name (input) into a search bar, and the app uses a RESTful API to send that search query to the backend. The backend then fetches relevant product information and sends it back to display.
    - **Login Form:** You enter your username and password (inputs). This data is sent via a RESTful API to the backend for authentication.
    - **Contact Form:** You fill in your name, email, and message (inputs). This information is bundled and sent to the backend via a RESTful API to be stored or emailed to the recipient.

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2. WebSocket Connections for Real-time Features

- **What it is:** Unlike RESTful APIs, which are like a waiter taking individual orders, WebSockets are like a continuous, open phone line between your app and the backend. This allows for instant, two-way communication without constantly asking for updates.
- **How input fields work:**

- **Chat Application:** You type a message (input) into a chat box. As soon as you hit enter, the message is instantly sent via a WebSocket connection to the backend, which then pushes it to other users in the chat room in real-time.
- **Live Sports Score Updates:** While not directly input, if there was a "commentary" input field, your comment would instantly appear for others via WebSockets.
- **Collaborative Document Editing:** Multiple users typing (inputs) into the same document. Each keystroke can be sent via WebSockets to update everyone else's view of the document in real-time.

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3. File Upload and Management APIs

- **What it is:** This allows your app to send files (like images, documents, or videos) to the backend for storage and later retrieval, or for the backend to manage these files (e.g., resizing images).
- **How input fields work:**
  - **Profile Picture Upload:** You click an "Upload Photo" button, which opens a file selection dialog. Once you select an image (input), the app uses a File Upload API to send that image file to the backend, where it's stored and associated with your profile.
  - **Document Submission:** For an online application, you might use an input field (often a button that triggers a file picker) to select and upload a resume or other documents.
  - **Media Sharing:** On social media, you select a video or image (input) to share, and the app uses these APIs to send the media to the server.

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4. Authentication and Authorization

- **What it is:**
  - **Authentication:** Verifying who you are (e.g., checking your username and password). It's like showing your ID at the entrance.
  - **Authorization:** Determining what you're allowed to do once you're inside (e.g., access certain pages or features). It's like the bouncer letting you into the VIP section.
- **How input fields work:**
  - **Login Form:** This is the primary input for **authentication**. You enter your **username/email** and **password**. The app sends these to the backend, which verifies your identity.
  - **Registration Form:** When creating a new account, you provide inputs like **desired username**, **password**, **email address**, etc. The backend processes this to create your new user profile and authenticate you.

- **Role-based Access:** While not direct input, after successful login (via input fields), the backend uses your authenticated identity to determine your **authorization** level, dictating which parts of the application you can see or interact with.

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5. Real-time Notifications and Updates

- **What it is:** This involves the backend proactively sending information to your app as soon as something new happens, without your app having to constantly ask for it. This often uses technologies like WebSockets (mentioned above) or other push mechanisms.
- **How input fields work:**
  - **Social Media Likes/Comments:** While you might not directly input to *receive* a notification, your activity (e.g., liking a post – which is often triggered by a click, not text input) on one user's frontend is sent to the backend. The backend then uses real-time notification mechanisms to send an update to the original post creator's app, notifying them of the like.
  - **Order Status Updates:** If you place an order (input via an order form), the backend tracks its status. When the status changes (e.g., "shipped"), the backend sends a real-time notification to your app to update you, possibly via a banner or a pop-up.
  - **New Message Alerts:** In a messaging app, when someone sends you a new message (their input), the backend immediately pushes a notification to your device.

Performance Optimization

Here's how each optimization point works and its indirect connection to input fields:

1. Code Splitting and Lazy Loading

- **What it is:** Imagine a huge textbook. Instead of making you download the entire book at once, **code splitting** breaks it into smaller chapters. **Lazy loading** means you only download a chapter when you actually turn to that page. For an app, this means only loading the code needed for the current screen or feature, and downloading other parts only when you try to use them. This makes the initial load time much faster.
- **How input fields work:**
  - **Conditional Forms:** If you have an input field that, when selected (e.g., a dropdown for "Account Type"), reveals a complex form specific to that type (e.g., "Business Account Details"), the code for that complex form can be lazy-loaded. You make an input selection, and *then* the necessary code is fetched, rather than downloading it all upfront.
  - **Feature Modules:** Imagine an app with an "Admin Panel" that only certain users can access. The code for this panel, which might contain many input fields for managing data, can be lazy-loaded. Only when an authorized user clicks an "Admin" button (an action triggered by an input, like a login that authenticates them) will the admin module's code be loaded.

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2. Caching Strategies for Frequently Accessed Data

- **What it is: Caching** is like having a copy of frequently used information closer to you so you don't have to go back to the main source every time. Think of it as your brain remembering a phone number you dial often, instead of looking it up in your contacts list every single time. For apps, this means storing data that's often requested (like product lists, user profiles, or common settings) either on your device or on a nearby server, so it loads instantly.
- **How input fields work:**
    - **Autocompletion/Search Suggestions:** When you start typing in a search bar (input), the suggestions that pop up are often pulled from a cache of frequently searched terms or product names. This makes the suggestions appear instantly without hitting the main database for every keystroke.
    - **Pre-filled Forms:** If you return to a form you've partially filled out, or a profile editing page, cached data might be used to pre-fill certain input fields, making the experience faster and more convenient.
    - **Dynamic Content Loading:** If you're Browse a product catalog, product details that have been viewed recently might be cached. Clicking on a product (not an input field, but a navigation action) would load its details instantly from the cache, rather than fetching them from the server again.

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3. Optimized Image and Video Loading

- **What it is:** Large images and videos can slow down an app significantly. **Optimization** here means making sure these media files are as small as possible without losing quality, loading them efficiently (e.g., loading low-quality versions first, or only loading them when they're visible on screen), and using modern formats.
- **How input fields work:**
    - **User Uploaded Content:** When you upload a profile picture or a video (via a file input field), the backend often applies image/video optimization techniques (resizing, compressing) *after* your input, to ensure they load quickly for other users viewing your profile.
    - **Gallery/Feed Scrolling:** While not direct input, as you scroll through a feed (an action driven by touch/mouse input), optimized loading ensures that images and videos appear smoothly without lag, as they are only loaded when they are about to become visible on your screen.

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4. Progressive Web App (PWA) Capabilities

- **What it is:** A **PWA** is like a website that acts and feels like a native mobile app. It can work offline, send push notifications, and be added to your home screen. This provides a more reliable and engaging user experience, even with patchy internet.

- **How input fields work:**
    - **Offline Form Submissions:** Imagine filling out a long survey or order form (using various input fields) while offline. A PWA can store your input locally and automatically sync it to the server once you're back online, preventing data loss and providing a seamless experience.
    - **Instant Loading from Home Screen:** When you launch a PWA from your home screen, it can load instantly because essential resources are cached. This means any initial input fields (like a login screen) appear without delay.

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5. Database Query Optimization for Large Datasets

- **What it is:** A **database query** is like asking a question to a giant library (the database) to find specific information. When you have a *very* large library, asking the question inefficiently can take a long time. **Query optimization** is about asking the right questions in the smartest way possible so the database can find the answer quickly, even with millions of records.
- **How input fields work:**
    - **Complex Search Filters:** If you use multiple input fields to apply advanced filters (e.g., "price range," "color," "brand") on an e-commerce site with millions of products, optimized database queries ensure that the results are returned almost instantly, rather than taking minutes.
    - **Reporting Tools:** For administrators or data analysts, input fields might be used to generate complex reports from massive datasets. Optimized queries are essential here to prevent the system from grinding to a halt while fetching the data for these reports.
    - **User Profile Retrieval:** When you log in (via input fields), the system needs to quickly retrieve your user profile and preferences from the database, even if there are millions of users. Optimized queries make this lookup fast.